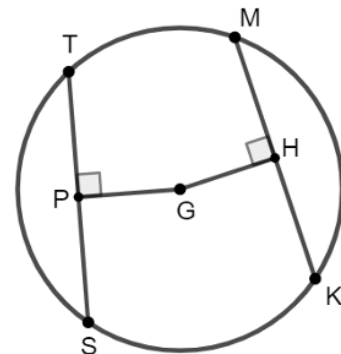


Geometry
Unit 12
Lesson 1 Practice

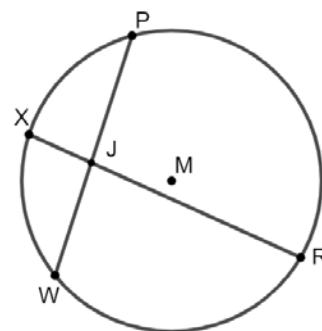
Name Answer Key
Date _____
Period _____

Circle G with points T, M, K , and S on the circle are shown with chords $\overline{TS}$ and $\overline{MK}$ equidistant from the center. $TS = (9x + 6)$, $MK = (11x - 10)$, and $m\angle TPG = m\angle MHG = 90^\circ$



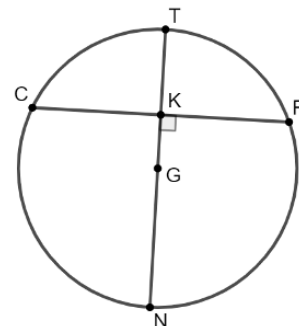
- Find the value of x .
 $9x + 6 = 11x - 10$
 $16 = 2x$
 $x = 8$
- What is the length of chord $\overline{MK}$?
 $MK = 11x - 10 = 11(8) - 10 = 78$

In Circle M , chords $\overline{RX}$ and $\overline{PW}$ intersect at point J . Points R, W, X , and P lie on Circle M . $RJ = (8x + 3)$, $JX = 4$, $PJ = (4x - 2)$, and $JW = 12$.



- Solve for the value of x .
 $12(4x - 2) = 4(8x + 3)$
 $48x - 24 = 32x + 12$
 $16x = 36$
 $x = 2.25$
- What is the length of line segment $\overline{RJ}$?
 $RJ = 8(2.25) + 3 = 21$
- Find the length of chord $\overline{PW}$.
 $PW = 12 + 4(2.25) - 2 = 19$

Circle G is shown where points C, T, R , and N are on the circle. The diameter, $\overline{NT}$, is a perpendicular bisector of chord $\overline{CR}$. $CK = (5x + 6)$ and $KR = (7x - 2)$.



- Calculate the value of x .
 $5x + 6 = 7x - 2$
 $-2x = -8$
 $x = 4$
- What is the length of chord $\overline{CR}$?
 $CR = 5(4) + 6 + 7(4) - 2$
 $CR = 20 + 6 + 28 - 2$
 $CR = 52$

Chord $\overline{CM}$ is drawn in Circle B so that it intersects the diameter $\overline{JP}$.

8. Given that $CM = 24$ and $TB = 5$, what is the length of $\overline{BP}$?

$$5^2 + 12^2 = BM^2$$

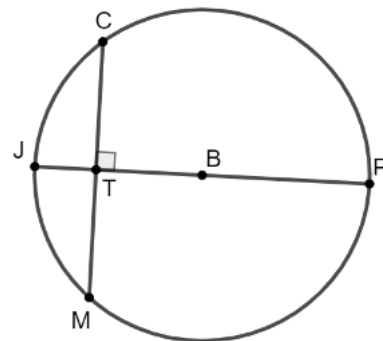
$$25 + 144 = BM^2$$

$$\sqrt{169} = \sqrt{(BM)^2}$$

$$13 = BM$$

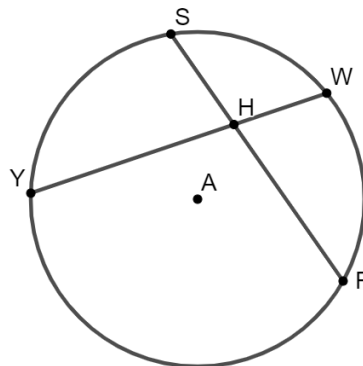
$\overline{BM}$ and $\overline{BP}$ are radii and therefore they are equal in length.

$$BM = BP = 13$$



Circle A is shown where points Y, S, W , and F lie on the circle. Chords $\overline{SF}$ and $\overline{YW}$ intersect at point H . $YH = (4x + 2)$, $SH = 9$, $HW = 8$, and $HF = (4x)$. Mia and Nazir are solving for x and their work is shown below.

Mia's Work	Nazir's Work
$8(4x + 2) = 9(4x)$ $32x + 2 = 36x$ $4x = 2$ $x = 0.5$	$8(4x + 2) = 9(4x)$ $32x + 16 = 36x$ $4x = 16$ $x = 4$



9. Whose work do you agree with? Justify your answer.

Nazir is correct. Mia did not distribute the 8 correctly to both terms in the parentheses.

In Circle H , points P, L, M , and S lie on the circle. Chord $\overline{PM}$ intersects the diameter $\overline{SL}$ at point G and $m\angle HGP = 90^\circ$.

10. If $PM = 32$ and $HS = 18$, what is the length of $\overline{GH}$? Round to the nearest hundredth if necessary.

$$18^2 = 16^2 + GH^2$$

$$324 = 256 + GH^2$$

$$\sqrt{68} = \sqrt{(GH)^2}$$

$$8.25 \approx GH$$

